A Proved Lift That Does Not Transfer: Split-Level Certificates of Non-Tree-Likeness, and Three Mechanisms of Failure

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Abstract

A quartet certificate refutes the four-point condition on four sequences with an exact integer comparison a proof kernel can recheck. It names four sequences; what a biologist reads is a *split*, one group of sequences against another. We promote the certificate to that level and report both halves of the result honestly.

The formal half succeeds. Because complete deletion gives every quadruple its own denominator, the Bandelt–Dress isolation index — a *minimum* over straddling quadruples — is not available without an incommensurable comparison, so we define the split certificate as a *logical conjunction* of per-quadruple integer comparisons instead. We then prove that two certified *incompatible* splits force a four-point violation, with the witness quadruple *constructed* from the four side-intersections rather than searched for (four theorems; the Lean module is 27/27 declarations with no `sorry` and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}).

The empirical half fails, and the failures are the contribution. Reaching a working simulated demonstration (20/20 specificity, 15/15 sensitivity across seven populations at five seeds) required three negative controls to fail first: the criterion had no noise floor and fired on a two-count margin in two thousand where an exact tree forces equality; no constant floor can separate, because saturation retains exactly $n-3$ splits at a floor $64\times$ the one that erases every other dataset; and the matched null was never matched, because a relaxed-clock simulator's divergence at fixed mutation rate has a coefficient of variation near 50% spanning 10–28×, so matching it *in expectation* is ill-posed however carefully implemented. Rejection sampling on each replicate's *realised* divergence repairs it.

The demonstration then does not transfer. At sixteen taxa on real viral populations the strict conjunction certifies 0, 2 and 4 splits where a tree has 13: every population reads as a tree, so its single agreement with a tree-like prior is vacuous. The mechanism is measured, not inferred — conjuncts for a balanced split number $\binom{n/2}{2}^2$, which is 36 at $n=8$ but 784 at $n=16$, and the closest-missing splits failed by *one* conjunct out of 91. An exception budget calibrated against the matched null recovers power and does not separate: all three populations, including the only tree-like control, stop reading as trees at a smaller budget than a genuine tree does. Any threshold between 13 and 24 would classify all three correctly, which on three populations with one negative control is fitting a cutoff to the answer, so we choose none.

We conclude that the binding constraint is not the statistic but the alignment admissibility gate, which leaves three of ten populations usable with a single negative control. One negative control cannot separate "this method false-positives" from "that population's prior is wrong." The next step for this line is more admissible negative controls, not another statistic.

1 Introduction and what is new

The four-point condition [1] is the tree-level null for a distance on a finite set: a metric is a tree metric exactly when, for all a, b, c, e,

$$d(a,b) + d(c,e) \;\le\; \max\big(d(a,c) + d(b,e),\; d(a,e) + d(b,c)\big). \tag{1}$$

A *quartet certificate* is four sequences, their six pairwise counts, and the one common denominator those counts share under complete deletion within the quadruple, such that (1) fails as an integer comparison. Prior work established such certificates, measured the cost of the alignment step that produces the columns, and showed that a certificate's margin is exactly twice one Bandelt–Dress isolation index, so that a certificate *is* the split-decomposition condition for a split system to fail to be a tree at the level of four taxa [9].

That identity invites an obvious promotion. A quadruple is not a biological object; a *split* is. This paper asks whether the certificate lifts, and reports three things that are new:

1. **A definition forced by exactness** (§2). The isolation index of a split is a minimum over the quadruples straddling it. Under complete deletion within the subset each such quadruple has a different denominator, so that minimum compares incommensurable quantities. Defining the split certificate as a *conjunction* of per-quadruple integer comparisons avoids this entirely and needs no common metric.

2. **A lift theorem** (§3). Two certified incompatible splits force a four-point violation, and the witness quadruple is constructed from the four side-intersections rather than searched for. Four machine-checked theorems, one of which proves a claim earlier editions of [9] asserted without proof.

3. **A measured non-transfer** (§§5–6). The method works on one simulated grid and has no power at the taxon count real data has. We give the scaling law responsible, the near-miss distribution that distinguishes "too strict" from "wrong," and the reason three populations cannot settle the question.

Along the way §4.3 reports a methodological finding with scope beyond this paper: a matched null calibrated on the *mean* of a nuisance parameter is not matched when the nuisance has a heavy-tailed realised distribution, and we give the measurement that detects this.

2 A conjunction, not a minimum

Write a quadruple's three pairing sums, over its own common columns, as $s_1 \le s_2 \le s_3$. Each of the three splits of the quadruple carries a Bandelt–Dress isolation index [2]

$$\alpha(A|B) \;=\; \tfrac{1}{2} \min_{a,a' \in A,\; b,b' \in B} \max\{\,0,\; \mathrm{cross}_1 - \mathrm{own},\; \mathrm{cross}_2 - \mathrm{own}\,\}, \tag{2}$$

which on a single quadruple reduces to $\frac{1}{2}\max\big(0, \max(\mathrm{cross}_1, \mathrm{cross}_2) - \mathrm{own}\big)$, and whose three values are therefore $\frac{1}{2}(s_3 - s_1)$, $\frac{1}{2}(s_3 - s_2)$ and 0. A split system is displayed by a tree exactly when it is pairwise compatible [3], so a split system containing two incompatible splits of positive index is the classical statement of non-tree-likeness.

The minimum in (2) is the obstruction. Complete deletion *within the subset* gives each straddling quadruple its own set of columns at which all four sequences carry a definite base, and hence its own denominator. Minimising a margin across quadruples with different denominators compares quantities that are not commensurable, and no choice of common denominator is available that does not reintroduce the columns complete deletion was there to remove.

Definition 2.1 (Split certificate)**.** Let θ be a non-negative rational. The split $A|B$ is *certified at floor* θ when, for **every** quadruple $\{a, a', b, b'\}$ with $a, a' \in A$ and $b, b' \in B$, the split's own pairing sum lies strictly below the larger crossing sum by more than θ times that quadruple's own denominator:

$$(\max(\text{cross}_1, \text{cross}_2) - \text{own}) \cdot \theta_{\text{den}} \;>\; \theta_{\text{num}} \cdot \text{common}.$$

Every conjunct is an integer comparison over its own denominator; nothing is averaged and no common metric is required. The split-level statement inherits its exactness from the quadruple level rather than paying for it. A split isolating a single taxon has no straddling quadruple, so an unguarded conjunction over its quadruples would be vacuously true; such splits carry no evidence and are excluded.

3 The lift theorem

Two splits are *incompatible* when all four of $A \cap C$, $A \cap D$, $B \cap C$, $B \cap D$ are non-empty. Choosing one taxon from each gives a quadruple on which the two splits induce two *different* pairings.

Theorem 3.1 (Two certified pairings force the third strictly largest)**.** *Let p and q be the pairing sums of two different splits of a quadruple and r the third. If $p < \max(q, r)$ and $q < \max(p, r)$, then $\max(p, q) < r$, and consequently $r \not\leq \max(p, q)$: the four-point condition fails on that quadruple.*

Proof. By $\max(p, q) < r \iff p < r \wedge q < r$ it suffices to establish both. From $p < \max(q, r)$ either $p < q$ or $p < r$; from $q < \max(p, r)$ either $q < p$ or $q < r$. The case $p < q$ with $q < p$ is impossible. If $p < q$ and $q < r$ then $p < r$ by transitivity and both hold. If $p < r$ and $q < p$ then $q < r$ and both hold. If $p < r$ and $q < r$ both hold directly. □

Corollary 3.2. *If two incompatible splits are both certified, then the four-point condition fails on the quadruple obtained by taking one taxon from each of the four side-intersections. The witness is* constructed*, not searched for.*

So a population-level claim — two groupings both certified, which no tree can display together — forces a quadruple-level certificate that a kernel can recheck. The following are machine-checked in `ANSE.GeneticaReticulataTree`:

declaration	content
`third_pairing_strictly_largest_of_two_certified`	Theorem 3.1
`fourPoint_violated_of_two_certified`	the same, as a four-point violation
`fourSet_splits_pairwise_incompatible`	no two of a four-set's three splits are compatible
`splitsCompatible_self`	a split never crosses itself

The module compiles 27 of 27 declarations with no `sorry` and no axioms beyond {`propext`, `Classical.choice`, `Quot.sound`}, checked by compiling and then reading `#print axioms` for every declaration, because `sorry` and a smuggled axiom both exit zero. `fourSet_splits_pairwise_incompatible` supplies a claim that editions 5 and 9 of [9, 10] asserted in prose without proof.

Remark 3.1. Theorem 3.1 uses no tree, no complex and no metric space — only three rationals. That is why the quantity survives per-subset evaluation, where a complex built over a global distance matrix cannot restrict to a subset without reintroducing the deleted columns.

4 The simulated demonstration, and the three controls that failed first

Seven populations at five seeds, each 2000 columns: a clonal strict-clock tree (D1), a clonal relaxed clock (D2), one planted recombinant under each clock (D3, D4), reciprocal mosaics (D5), gap blocks (D6), and a saturated population (D7). D3, D4 and D5 carry reticulation; the rest do not and are the controls. We report the three control failures because each located something the original question did not.

4.1 The criterion had no noise floor

With $\theta = 0$, Definition 2.1 is own $<$ max(cross), and the negative control failed 0 of 5: the clonal strict-clock population certified 8 splits where its tree has 5, and 6 of the resulting pairs were incompatible. The first exhibit shows why. Its pairing sums were $[66, 110, 112]$ over a denominator of 2000 — and on an *exact* tree metric the two crossing sums are **equal**. Their observed difference is sampling noise of order $\sqrt{L}$, and the sign of that noise is what the criterion was reading: the split with 110 was certified on a two-count margin in two thousand.

Remark 4.1. A pre-registered prediction also failed to be a test. It asked whether a planted recombinant was implicated, accepting it either in a witness quadruple or anywhere in the incompatible pair's splits — and a split's two sides cover *every* taxon, so the second disjunct is true whenever any incompatible pair exists. Only the witness quadruple, four taxa of nine, carries information. The forced form recorded a pass; the repaired form is what §4.4 reports.

4.2 No constant floor can separate

Sweeping θ over ten values, no floor separates the reticulate populations from the controls. Tree-readings per five seeds, where a tree-reading is correct for D1, D2, D6, D7 and a miss for D3, D4, D5:

θ	D1	D2	D6	D7	D3	D4	D5
0.00100	5/5	5/5	1/5	1/5	0/5	0/5	0/5
0.00400	5/5	5/5	5/5	3/5	2/5	2/5	0/5
0.01600	5/5	5/5	5/5	5/5	5/5	5/5	5/5

The saturated population retains exactly $n - 3 = 5$ certified splits at $\theta = 0.064$, sixty-four times the floor that erases every other dataset, because saturation inflates margins *in rate terms*. A single global rate is the wrong functional form.

4.3 The matched null was never matched

Calibrating the floor per population against a clonal null matched on divergence fixed specificity but reported a null whose divergence missed the population's by 77%, and only on the two relaxed-clock datasets. Two defects compounded.

First, the tuner bisected a *single realisation*. The relaxed-clock simulator draws its per-branch rate multipliers from the same pseudorandom stream as its mutations, and the mutation step consumes a number of draws that depends on the rate; changing the rate therefore shifts the stream and changes every downstream multiplier. A single realisation's divergence is consequently a chaotic, non-monotone function of the rate, and bisecting it wanders rather than converging. The strict-clock simulator has nothing downstream of its mutations, which is exactly why only relaxed-clock nulls were affected.

Averaging the objective over five seeds barely helped — worst relative error 147.5% against 143.7% — so we measured the assumption underneath. At *fixed* rate, over 200 realisations:

rate	mean divergence	CV	min	max/min
0.001	0.00658	57.6%	0.00088	28.4×
0.004	0.02677	49.1%	0.00521	13.6×
0.016	0.09299	48.9%	0.02272	10.1×

Each lineage's rate is its parent's times $\exp(\sigma u)$, so rate multipliers *compound* across generations and their log-variance grows with depth. The distributions at rates 0.001 and 0.004 overlap heavily. This is the second and deeper defect: the floor is a **maximum** over replicates, so when the realised divergences of those replicates differ by up to 28×, the maximum is set by whichever replicate drew a fast-rate tree. The null was inflated, hence conservative, but not matched.

Remark 4.2 (A finding with scope beyond this paper). Matching a null on the *mean* of a nuisance parameter is not matching it when the nuisance has a heavy-tailed realised distribution, and averaging more replicates does not repair this: averaging shrinks the standard error of the *tuning*, not the spread of the *replicates* the statistic is computed on. The diagnostic is cheap and we recommend it generally — measure the across-realisation coefficient of variation of the quantity being matched, and if it is comparable to the range being matched over, expectation-matching is ill-posed.

4.4 Rejection sampling on the realised nuisance

A replicate counts only if *its own* realised divergence is within 10% of the population's. With that, four pre-registered predictions hold at full count: every non-reticulate population reads as a tree (20/20); every reticulate population does not (15/15); the reticulate ones still certify something, so an empty system cannot masquerade as a verdict (15/15); and a planted recombinant appears in a witness quadruple (15/15).

dataset	reticulate	floor	certified	$n-3$	trees	acceptance
D1 clonal, strict clock	no	0.00170	**5.0**	5	5/5	72.5%
D2 clonal, relaxed clock	no	0.00120	4.4	5	5/5	17.1%
D6 gap blocks	no	0.00339	4.0	5	5/5	52.5%
D7 saturated	no	**0.04000**	**5.0**	5	5/5	100.0%
D3 one recombinant	**yes**	0.00160	9.0	6	0/5	54.7%
D4 relaxed + recombinant	**yes**	0.00190	7.4	6	0/5	15.5%
D5 reciprocal mosaics	**yes**	0.00170	10.6	7	0/5	51.5%

D1 and D7 recover exactly $n-3=5$ certified splits, their true internal split count, with no incompatible pairs; D7's floor is 23× D1's, which is the divergence dependence no constant rate could express. Acceptance rates are reported because they matter: a population that cannot be matched receives no floor and no verdict rather than a floor from an unmatched null. Across all 350 dataset–seed–floor combinations, every constructed witness really violates the four-point condition.

5 Non-transfer at sixteen taxa

The simulated grid has 8–10 taxa. Real viral populations, filtered by pre-registered quality criteria and aligned to a chosen reference, have 16. The same pipeline that produced the quartet results of [9] — the same reference-anchored alignment, the same two references, and the same *measured* admissibility threshold above which a population is reported blocked rather than interpreted — admits three of ten populations.

5.1 Making the computation possible

Certifying every split of 16 taxa requires about 39.7 million straddling quadruple evaluations. There are only $\binom{16}{4} = 1820$ distinct quadruples, each reused about 21 800 times, so we precompute their counts once (about 1.3×10^7 column comparisons) and the rest are table lookups; done directly it is about 2.8×10^{11}. The table must identify which of the three canonical pairing sums a split induces, which it does by asking whom the smallest index is paired with — index arithmetic that is easy to get subtly wrong and invisible in aggregate output, so it is checked exhaustively against directly computed counts for every quadruple and every 2+2 split, and the refactor is confirmed to reproduce the simulated artifact exactly.

5.2 Zero power

A 16-taxon tree has $n - 3 = 13$ internal splits.

population	prior	certified splits	reads as
Coxsackievirus A6	reticulate	**0** / 32751	tree
Hepatitis B	reticulate	**2** / 32751	tree
JC polyomavirus	tree-like	**4** / 32751	tree

Every population reads as a tree. The single agreement with a tree-like prior is therefore *vacuous*: a method that always answers "tree" agrees with every tree-like prior for no reason, and reporting 1/3 here would be reporting an artefact of having no power.

5.3 The mechanism, measured

Definition 2.1 requires every straddling quadruple to clear the floor, and the number of conjuncts for a split with k taxa on one side is $\binom{k}{2}\binom{n-k}{2}$, maximised at $\binom{n/2}{2}^2$: **36 at $n = 8$, but 784 at $n = 16$.** A conjunction over hundreds of noisy comparisons dies on a single dissent.

The near-miss distribution separates "the method is wrong about which splits exist" from "the method is too strict to certify one," and these demand opposite conclusions:

population	closest miss	conjunction length
Coxsackievirus A6	**1** failing conjunct	91
Hepatitis B	2 failing conjuncts	91
JC polyomavirus	11 failing conjuncts	234

One dissenting quadruple out of ninety-one. The rule is not wrong about the splits; it is too strict to certify one at this conjunction length, and it succeeded on the simulated grid only because eight taxa made the requirement roughly twenty times weaker. We had stated the simulated scope as "one parameter grid, 8–10 taxa" and treated that as adequate; it named parameters without identifying which one the method's difficulty scales in, and taxon count was that one.

6 An exception budget, and why three populations cannot settle it

Allowing at most k dissenting conjuncts keeps the condition *exact and countable* — "at most k of these integer comparisons fail" is as checkable as "none does" — so it is a weaker claim, not a vaguer one. What it must never be is a chosen number. We calibrate k as the largest value at

which *every* matched null replicate still reads as a tree: one step beyond, a genuine clonal tree at the population's own divergence would be called reticulate. This gives $k = 23$, with the null first breaking at 24.

That recovers power and produces a false positive: Coxsackievirus A6 certifies 8 splits and reads as a non-tree (correct), Hepatitis B certifies 18 and reads as a non-tree (correct), and JC polyomavirus certifies 6 and reads as a **non-tree**, which its prior says is wrong.

A threshold invites fitting, so we replace it with a comparison carrying no cutoff: the budget at which the population stops reading as a tree, against the budget at which its *own* matched null does.

population	prior	population breaks at	null breaks at	difference
Hepatitis B	reticulate	3	24	−21
Coxsackievirus A6	reticulate	9	24	−15
JC polyomavirus	**tree-like**	13	24	**−11**

All three break earlier than a tree, *including the negative control*. The ordering is in the right direction — the reticulate populations break earliest, the tree-like one is intermediate, a genuine tree latest — but the statistic does not separate. Any threshold between 13 and 24 would classify all three correctly, and selecting such a threshold from three populations with one negative control is fitting a cutoff to the answer. We select none and report the comparison as it stands. Both alignment references agree on every verdict, so none of this is an alignment artefact.

Remark 6.1. One negative control cannot distinguish "this method has a false-positive problem" from "this population's tree-like prior is wrong." These priors are priors: [9] already records that one of the negative-control families has documented within-type recombination [7], and a prior drawn from a literature consensus is not a label. With a single control the two explanations are observationally identical, and no additional statistic computed on the same three populations can separate them.

7 Related work

Split decomposition [2] decomposes a metric into a totally split-decomposable part and a split-prime residue, and the isolation index of (2) is its central quantity; the splits-equivalence theorem relating pairwise compatibility to displayability by a tree is standard [3]. Our contribution at this level is not the index but the observation that complete deletion within the subset makes its defining minimum incommensurable, and that a conjunction recovers an exact substitute.

Statistics for non-tree-likeness computed by aggregating over quadruples include δ plots [4], the pairwise homoplasy index [5], and quartet-based summary tests for introgression [6]. These average or count; the certificate exhibits. The distinction is the point of the programme rather than a claim of superior power, and [9] reports the quartet certificate tying the pairwise homoplasy index on power under matched nulls.

The relationship between split decomposability and the topology of Vietoris–Rips complexes is actively studied: Gómez [8] investigates Vietoris–Rips complexes of split-decomposable spaces, motivated by the fact that totally decomposable spaces generalise metric trees and are therefore attractive candidates for efficiently computable persistent homology. We note this because a companion result closed a distance-magnitude Rips filtration on measured grounds for this task, and that closure concerned a specific implementation — a global complex cannot use per-subset complete deletion, and a magnitude-driven filtration tracks divergence rather than the index a tree forces to zero — and not the mathematical direction, which [8] shows to be live. We have read only the abstract of [8] and cite it for that framing alone.

8 Limitations

The formal content of §3 is proved and the empirical content is one simulated grid. Specifically:

- **The lift is unvalidated on real data.** It works at 8–10 simulated taxa and does not transfer to 16 real taxa. The strict form has no power there and the calibrated form cannot be shown to separate.
- **Three populations, one negative control.** This cannot establish specificity, and we do not claim it does. The admissibility gate is what limits the sample, not the availability of sequences.
- **Theorem 3.1 is about three rationals.** It says nothing about whether a certified split corresponds to a biological event; that inference remains the reader's, informed by the exhibited sequences.
- **The exception budget weakens the claim.** "At most k conjuncts fail" is exact but is a weaker statement than the strict conjunction, and its calibration depends on the matched null being correctly matched — which §4.3 shows is a non-trivial requirement that an earlier version of this apparatus did not meet.
- **Priors are not labels.** The literature expectations used as ground truth are consensus readings, one of which is known to be weak.

What would settle it is more admissible negative controls — by a better alignment step whose error is measured at the loads real data presents, or by pre-aligned data of independent provenance — and not another statistic computed on these three populations.

9 Reproduction

All numbers in this paper are produced by committed code and stored artifacts, and no result is hand-entered. The Lean module is verified by compiling and reading `#print axioms` for every declaration. The simulated experiment, the parameter sweeps, the nuisance-variability measurement and the real-data run are separate subcommands writing JSON artifacts; the refactor that made the 16-taxon computation possible is confirmed behaviour-preserving against the committed simulated artifact. The accompanying software release records the test counts, the clippy warning count, and the guard outputs for the edition that produced these figures.

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